

Data Structures — Assignment Solutions

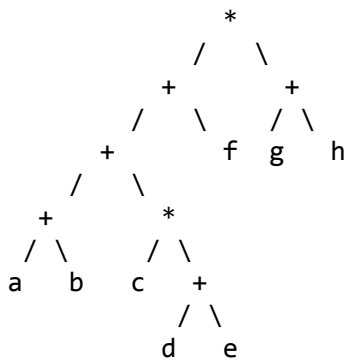
Question 1: Prefix and Postfix Conversion

Given infix expression:

```
((a + b) + c * (d + e) + f) * (g + h)
```

Step 1: Expression Tree

Using operator precedence (* before +) and left-to-right associativity for the additions, the expression parses into the following binary tree structure (each node is an operator, leaves are operands):



Step 2: Prefix Expression (Pre-order traversal: Root, Left, Right)

```
* + + + a b * c + d e f + g h
```

Step 3: Postfix Expression (Post-order traversal: Left, Right, Root)

```
a b + c d e + * + f + g h + *
```

Verification

Both expressions contain the same 7 operators as the original infix expression: +, +, *, +, +, +, * — confirming the conversions are consistent.

Question 2: Recursive Binary Search

Array (sorted, n = 11 elements, indices 0–10):

```
[23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 88]
```

Key to search: 88

C Function (Recursive Binary Search)

```
#include <stdio.h>
```

```

int binarySearch(int arr[], int low, int high, int key) {
    if (low > high) {
        return -1;                // key not found
    }

    int mid = (low + high) / 2;

    if (arr[mid] == key) {
        return mid;                // key found
    }
    else if (key < arr[mid]) {
        return binarySearch(arr, low, mid - 1, key);    // search left half
    }
    else {
        return binarySearch(arr, mid + 1, high, key);    // search right half
    }
}

int main() {
    int arr[] = {23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 88};
    int n = 11;
    int key = 88;

    int result = binarySearch(arr, 0, n - 1, key);

    if (result != -1)
        printf("Key %d found at index %d\n", key, result);
    else
        printf("Key %d not found\n", key);

    return 0;
}

```

Trace of Execution (key = 88)

Step	low	high	mid	arr[mid]	Comparison	Action
1	0	10	5	58	88 > 58	search right half → low = mid + 1 = 6
2	6	10	8	68	88 > 68	search right half → low = mid + 1 = 9
3	9	10	9	73	88 > 73	search right half → low = mid + 1 = 10
4	10	10	10	88	88 == 88	Key found at index 10

Result: The key 88 is found at index 10 after 4 recursive calls.